FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

BIOLOGY

Past Paper 2025 • Paper Code: 2HA-I 25006 — 40061 (Version B) / 42061 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

BIOLOGY HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 40061 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. A person experiencing severe stress has an increased heart rate and blood pressure. Which hormone is primarily responsible for this?

(A) Adrenaline
(B) Insulin
(C) Thyroxin
(D) Oxytocin

Q2. When you inhale, the diphragm:

(A) Relaxes and moves upward
(B) Relaxes and moves downward
(C) Contracts and moves upward
(D) Contracts and moves downward

Q3. A school of fish swims together in a coordinated manner, with each fish following the movement of the others. It is an example of:

(A) Social behavior
(B) Innate behavior
(C) Learned behavior
(D) Migratory behavior

Q4. A couple has a child with cystic fibrosis: An autosomal disorder. If both parents are carrier of the disease but do not show symptoms. What is the probability that their next child will have cystic fibrosis?

(A) 25%
(B) 50%
(C) 75%
(D) 100%

Q5. Which biotechnology technique is used to produce human insulin on large scale?

(A) Gene therapy
(B) Genetic engineering
(C) Cloning
(D) Fermentation

Q6. Which is an example of negative feedback mechanism?

(A) Insulin release and increased blood glucose
(B) Adrenaline release and increased heart rate
(C) ADH release and increased water reabsorption
(D) Thyroxin release and increased metabolic rate

Q7. Put the following statement in proper order to describe muscle contraction: 1 Signal sent by CNS via motorneuron; 2 Sarcomere Shortens; 3 Generation of action potential in the sarcolemma; 4 Release of Ca+2 from sarcoplasmic reticulum

(A) 4→2→1→3
(B) 1→2→3→4
(C) 4→3→2→1
(D) 1→3→4→2

Q8. A population of bacteria develops resistance to antibiotics over time. Which evolutionary mechanism is most likely responsible for this adaptation?

(A) Genetic drift
(B) Natural selection
(C) Mutation
(D) Gene flow

Q9. In the following graph of action potential in a neuron, X shows. — [A graph of membrane potential (mV) vs time is printed in the source scan; X points to the rising peak of the action potential curve]

(A) Hyperpolarization
(B) Repolarization
(C) Depolarization
(D) Polarization

Q10. Spermatogenesis in human male reproductive system takes place in:

(A) Seminiferous tubules
(B) Epididymis
(C) Vas deference
(D) Urethra

Q11. During which stage of human development does puberty occur:

(A) Infancy
(B) Childhood
(C) Adolescence
(D) Adulthood

Q12. Identical twins result from the fertilization of:

(A) 1 ovum by 1 sperm
(B) 1 ovum by 2 sperm
(C) 2 ova by 2 sperm
(D) 2 ova by 1 sperm

Q13. Which DNA strand can base pair with the shown DNA strand? — [The shown DNA strand ATGCTA is printed in the source scan]

(A) TACGAT
(B) UACGAU
(C) ATGCTA
(D) AUGCUA

Q14. In the food web shown, which of the organisms X, Y or Z. Is herbivore: — [A food-web diagram (Grass → Grasshopper; X, Y, Z feeding links) is printed in the source scan]

(A) Z
(B) Both X, Y
(C) Y
(D) X

Q15. In the following diagram which region represents one sarcomere? — [A sarcomere/ myofibril diagram with regions labelled 1–4 is printed in the source scan]

(A) 1
(B) 2
(C) 3
(D) 4

Q16. Which option most accurately describes the sequence of air flow in human respiratory system. 1. Pharynx 2. Bronchus 3. Trachea 4. Alveoli 5. Bronchiole

(A) 1,2,3,4,5
(B) 1,3,2,5,4
(C) 5,4,3,2,1
(D) 1,5,2,3,4

Q17. A backcross is made when a hybrid is crossed with its:

(A) Siblings
(B) Offspring
(C) Parents
(D) Other species

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 25006 (B)

Q2. (Marks 14x3=42) Q.2 Answer the following parts briefly.

(i) [03] Describe briefly the adaptation of the human body to extreme temperature. — OR (03) — What are extraembryonic membranes? Write the name and functions of any three.

(ii) [03] Explain briefly the mechanism of water and electrolyte balance in the human body. — OR (03) — Describe briefly the role of telomeres in aging.

(iii) [03] Explain briefly the role of neural crest in embryonic development. — OR (03) — A couple has child with sickle cell anaemia, an autosomal disorder. If one parent is a carrier and the other is normal, what is the probability that their next child will be a carrier?

(iv) [03] What is allometric growth? Give an example. — OR (03) — Describe briefly the inheritance pattern of a sex influenced trait.

(v) [03] Analyze the following pedigree for a genetic disorder. What is the probability that child 3 will be affected? — [A pedigree chart is printed in the source scan] — OR (03) — What is the primary role of restriction enzymes in genetic engineering?

(vi) [03] A couple has a child with a genetic disorder caused by a mutation in Mitochondrial gene. Who is more likely to pass on the mutation to their offsprings? Explain briefly. — OR (03) — Write three major steps involved in DNA sequences.

(vii) [03] Explain briefly the concept of Neutral selection. (Genetic drift). — OR (03) — What is the purpose of vector and DNA ligase in DNA cloning?

(viii) [03] Describe briefly the structure of uterus. — OR (03) — A patient is diagnosed with a genetic disorder caused by a defective gene. How can gene therapy be used to treat this disorder? Explain briefly.

(ix) [03] What is the primary function of respiratory system in humans? How is it helpful for the body? — OR (03) — How might changes in gene expression contribute to cancer development?

(x) [03] What is the significance of molecular biology in supporting evolution? — OR (03) — What is the role of fallopian tubes in female reproductive system?

(xi) [03] How do microorganisms play a crucial role in sewage treatment? — OR (03) — List down the conditions required to meet the Hardy Weinberg theorem.

(xii) [03] Explain briefly the difference between sensory and motor neurons. — OR (03) — What is the significance of the hypothalamus in the endocrine system?

(xiii) [03] Explain briefly integrated disease management. — OR (03) — Explain briefly the concept of climax community and its significance.

(xiv) [03] Explain briefly the concept of ozone depletion and its significance. — OR (03) — What is the role of diaphragm in respiration?

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 4+2) Q.3 Explain the XX-XY mechanism of sex determination in man. Why males are known to be heterogametic? — OR (4+2) — Describe the mechanisms involved in the transport of carbon dioxide from the tissues to lungs in human blood. Also name the process. Explain what would happen if no carbonic anhydrase is present in red blood cells?

Q4. (Marks 4+3) Q.4 Describe the steps involved in gene amplification through PCR. Also enlist some of its applications. — OR (1+6) — What are ecological pyramids? Draw and explain pyramids of energy, Biomass and number.

Q5. (Marks 1+6) Q.5 What is meant by action potential? Graph shows action potential in a neuron. Identify and explain the processes labelled as X, Y, Z. — [An action-potential graph (voltage vs time, labels X, Y, Z) is printed in the source scan] — OR (5+1+1) — Describe the events of the process of DNA replication. Interpret how DNA conserve one strands during this process? Identify the labelled parts A and B. — [A DNA replication diagram (labels A, B) is printed in the source scan]

Q6. (Marks 1+2+3) Q.6 How is animal society defined? What are its benefits? Explain the concept of Altruism with the example of Honey bee hive. — OR (2+2+2) — The given figure shows three types of muscles for movement in animals. Identify and compare the properties of these muscles. — [Three muscle diagrams labelled A, B, C are printed in the source scan]

VERSION D

BIOLOGY HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes	**Maximum Marks:** 17	**Paper Code:** 42061 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. A marathon runner has high sodium levels in his blood. What helps restore the balance?

(A) ADH
(B) Aldosterone
(C) Increased sweating
(D) More urine production

Q2. The most suitable medicine to treat hypothyroidism is:

(A) TSH (Thyroid Stimulating Hormone)
(B) Adrenaline
(C) Insulin
(D) Synthetic thyroxin (T4)

Q3. A flock of birds migrates to warmer climates every winter. It is an example of:

(A) Innate behavior
(B) Learned behavior
(C) Social behavior
(D) Imprinting

Q4. A man with blood type AB marries a woman with blood type O. The most probable blood type of their children is:

(A) A or B
(B) A, B or AB
(C) A, B or O
(D) Only AB

Q5. Fossil records show that a particular species of bird evolved from a lizard like ancestor over millions of year. Which mechanism of evolution is most likely responsible for this transition?

(A) Genetic drift
(B) Natural selection
(C) Mutation
(D) Gene flow

Q6. A farmer wants to grow crops that are resistant to a specific pest. The most suitable biotechnology for it would be:

(A) Genetic Engineering
(B) Gene Therapy
(C) Cloning
(D) Bioremediation

Q7. What is the primary goal of homeostasis?

(A) Regulate emotional responses
(B) Coordinate movement
(C) Defend against pathogen
(D) Maintain stable internal environment

Q8. Following is the diagram of a skeletal muscle band. What labelled part shortens during muscles contraction? — [A skeletal-muscle band diagram with labels i, ii, iii is printed in the source scan]

(A) i
(B) i, ii
(C) iii
(D) ii, iii

Q9. Which of the following conditions best describes resting membrane potential?

(A) ++++ outside / ++++ inside
(B) ++++ outside / ---- inside
(C) + - + - outside / + - + - inside
(D) ---- outside / ++++ inside

Q10. In lumen of seminiferous tubules liquid medium, nourishments and protection to sperms is provided by:

(A) Sertoli cells
(B) Germinal epithelium
(C) Interstitial cells
(D) Seminal vesicles

Q11. The function of human chorionic gonadotropin hCG is:

(A) Stimulates the uterus to secrete estrogen
(B) Maintain corpus luteum to secrete progesterone
(C) Stimulates placenta to secrete progesterone
(D) Stimulates hypothalamus to secrete hormones

Q12. Which of the following consists of both fetal and maternal tissues?

(A) Umbilical Cord
(B) Placenta
(C) Amnion
(D) Allantois

Q13. A chemical causes the change in sequence of nucleotide as shown, is an example of: — [Nucleotide sequences ATGAAA → ATTAAA are printed in the source scan]

(A) Translocation
(B) Frameshift mutation
(C) Point mutation
(D) Inversion

Q14. In the shown diagram less energy is available in the second level because: — [An energy-pyramid diagram is printed in the source scan]

(A) There is more food than at the first level
(B) Energy from first level was given off as heat
(C) The organism at top does not need much energy
(D) Producers use less energy than consumers

Q15. When ligaments are stretched or torn, it can lead to a condition called:

(A) Sprain
(B) Dislocation
(C) Fracture
(D) Tension

Q16. Which response would help compensate for low O2 levels at high altitudes?

(A) Increased breathing rate
(B) Decreased heart rate
(C) Constriction of blood vessels
(D) Decreased sweating

Q17. Animal husbandry is a branch of science that deals with care of:

(A) Disease
(B) Fruits
(C) Livestock
(D) Vaccination

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 25006 (D)

Q2. (Marks 14x3=42) Q.2 Answer the following parts briefly.

(i) [03] Briefly explain the concept of set point and its significance in homeostasis. — OR (03) — Describe the process of neurulation and its significance.

(ii) [03] Describe the role of kidneys in regulating fluid balance. — OR (03) — What is the significance of blastula stage in embryonic development?

(iii) [03] Name the three germ layers. Enlist one tissues / organs developed by each germ layer. — OR (03) — A woman has a recessive genetic disorder that is linked to the X chromosome. If she has two sons. What is the probability that they will inherit the disorder?

(iv) [03] Briefly describe meroblastic cleavage with an example. — OR (1+2) — In the following human pedigree, what is the probability of child labelled as 'Z' will be affected? — [A pedigree chart is printed in the source scan]

(v) [03] Briefly explain the role of epistasis in shaping phenotypes. — OR (03) — A forensic scientist needs to analyse DNA sample from a crime scene. What biotechnological technique can be used to analyse the sample? Enumerate main steps.

(vi) [03] How does gel electrophoresis separate DNA fragments? — OR (03) — Describe the regulation of breathing by the nervous system.

(vii) [03] Enumerate the main steps for creating a recombinant DNA molecule. — OR (03) — What is the role of haemoglobin in transporting oxygen in the blood?

(viii) [03] A community is affected by a water borne disease. How biological treatment methods can be used to treat the sewage? Explain briefly. — OR (03) — What is the function of ovaries in the female reproductive system?

(ix) [03] What is the role of myelin sheath in nerve conduction? — OR (03) — Describe the role of microbes in energy production.

(x) [03] How do histone proteins contribute to the formation of nucleosomes? — OR (03) — How does comparative anatomy provide evidence for evolution?

(xi) [03] Briefly explain the concept of Natural selection. — OR (03) — What are the three main causes and effects of climate change?

(xii) [03] Describe the process of convergent evolution with an example. — OR (03) — Briefly explain the concept of pioneer species and their role in succession.

(xiii) [03] Describe the mechanism of action of insulin on target cells. — OR (03) — Explain briefly the concept of Neutral selection. (Genetic drift)

(xiv) [03] What is the purpose of marker gene like antibiotic resistant gene in plasmid? — OR (03) — What is the role of estrogen and progesterone in female reproductive physiology?

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 1+4+1) Q.3 The figure shows respiratory system in man. Name and explain the processes happening in figures 'a' and 'b'. Lungs are spongy in nature. They never pull air or push it out. Then how does breathing takes place? — [Two respiratory figures (a) and (b) are printed in the source scan] — OR (2+4) — Identify the labelled parts A-D. Explain synaptic transmission of nerve impulse as well. — [A synapse diagram with labels A–D is printed in the source scan]

Q4. (Marks 2+5) Q.4 Identify the labelled parts from A – D. How cross bridges are formed during muscle contraction? Explain. — [A muscle-contraction diagram with labels A–D is printed in the source scan] — OR (5+2) — Associate multiple alleles with ABO blood group system. What blood type can a person with blood group A receive?

Q5. (Marks 5+2) Q.5 Explain the mechanism of transcription. Why the length of transcribed mRNA shortens as it enters the cytoplasm for translation? — OR (4+3) — Explain tissue culture. How this technique is different in plants and animals?

Q6. (Marks 2+4) Q.6 Describe innate behaviour. Explain with examples taxis shown by unicellular organism and tropism shown by plants. — OR (3+3) — Draw line diagram of nitrogen cycle. Describe the role of microorganism in nitrogen cycle.